FUNNOW Hotel AI Reporting Pipeline

Overview

This project automates FUNNOW Group's weekly hotel reporting. The Python pipeline ingests CSV data, evaluates hotel performance deterministically, uses an LLM to generate personalized account-management summaries, and securely dispatches them through Gmail SMTP.

Tool Selection

- **Python & Pandas:** Chosen for robust data manipulation, allowing for portfolio-wide calculations before any LLM text generation occurs.
- **Groq API** (***qwen/qwen3.8-27b***)**:** Selected for fast, cost-effective inference. The model handles natural language formatting, while the business logic remains mathematically auditable.
- **SMTP & *python-dotenv*:** Utilizes Python's built-in libraries for reliable email dispatch while keeping API keys and credentials secure outside the codebase.

Performance Thresholds

The pipeline dynamically calculates the median for both revenue and order volume based on the current dataset. The median was chosen over the mean to create a relative benchmark which is defensive to massive outliers.

To classify a hotel, the system requires alignment across both metrics. Properties meeting or exceeding both medians are tagged as "Performing Well", while those falling below both are tagged as "Needs Improvement". Any property with mixed metrics (e.g., high revenue but low orders) is tagged as "Mixed Results". This dual-metric requirement prevents one inflated metric from masking a weakness in the other.

AI Message Design

The prompt is tightly scoped, assigning the LLM a "FUNNOW Account Manager" persona and injecting only deterministically calculated facts to prevent hallucination.

To drive business value, the prompt combines the status with actionable advice such that Needs Improvement triggers a suggestion to run a FUNNOW flash promotion to boost visibility, while Performing Well triggers encouragement to keep weekend inventory open. The output is strictly limited to 3-4 sentence.